
Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 22 tissue sections from four technologies (336 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, implicit neural-field and imputation baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, so we assembled 10 independent specimens across five organs and two donors’ species. At that level the advantage survives family-wise correction against 3 baselines (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN); Holm-corrected $p \leq 0.020$, winning on at least 9/10 specimens), including one graph model. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r , more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.002, against 0.014 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the strongest graph baseline the separation is still unresolved (STAGATE-style: 7/10 specimens, $p = 0.43$), so the method is distinguishable from most of this literature but not from all of it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.

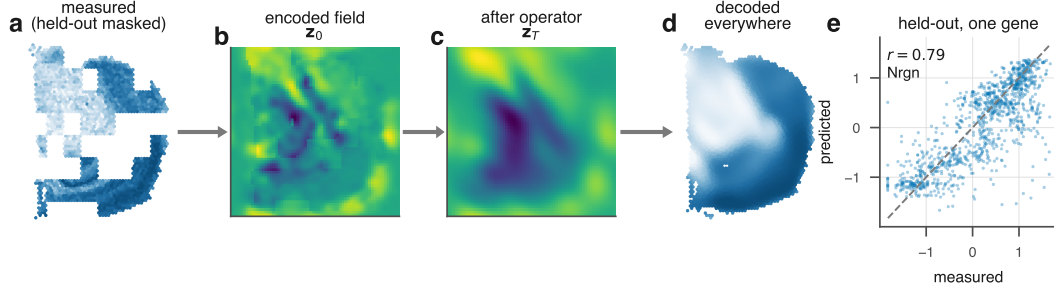


Figure 1: **Neural Reaction-Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction-diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction-diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction-Diffusion Operators (NRDO)**, which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction-diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction-diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions ??–1); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts our perturbation result.

66 2 Method

67 A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized
68 to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode–
69 evolve–decode.

70 **Spatial encoder.** Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al.,
71 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement
72 to a per-channel gain, so the layer depends on geometry rather than node identity. Features are
73 splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and
74 an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral
75 blocks [Li et al., 2021] refine the field.

76 **The reaction–diffusion operator.** The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

77 with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so
78 definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as
79 1×1 convolutions with a tanh-bounded output.

80 Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t =$
81 $O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the
82 Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

83 Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-
84 step map that is second-order accurate (Proposition ??). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2)
85 lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean
86 exactly, with no CFL restriction (Proposition ??). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this
87 holds along the entire optimization trajectory — no step size couples to learned parameters, and no
88 projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-
89 mean component (Proposition ??). Both are proved in Appendix ?? and checked numerically in
90 Table ??; Section ?? measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths
91 per unit pseudo-time, convertible to microns via the stored coordinate scale.

92 **Decoder.** Predictions come from bilinear interpolation of the evolved field at any continuous co-
93 ordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could
94 memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation
95 uninterpretable.

96 **Objectives.** The objective combines reconstruction (MSE plus a per-gene Pearson term, since the
97 spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and
98 physics terms. The conventional physics-informed penalty—the PDE residual along the model’s
99 own trajectory—turns out to be uninformative for an exactly integrated operator.

100 Proposition 1 makes this precise: the residual along the model’s own trajectory is $\frac{\Delta t}{2} \|[\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta]\mathbf{z}\| +$
101 $O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing
102 it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward
103 operators whose diffusive and reactive parts commute (Appendix A). We instead require that the
104 *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

105 the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$
106 and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

107 **Interpretability.** Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_k$ with
 108 $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned
 109 operator — a falsifiable read-out of the dynamics it has inferred (Appendix ??).

110 3 Experimental setup

111 **Data.** We use 12 public datasets spanning four spatial technologies (23 tissue sections, 984,112
 112 measured locations; Fig. ??), plus one non-spatial perturbation screen (111,659 dissociated cells)
 113 used only in Section ?? . Raw reads are never used: we consume vendor count matrices and record a
 114 SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization,
 115 log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so
 116 the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates
 117 are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned
 118 diffusion coefficient meaningless.

119 **Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the en-
 120 coder and every model predicts expression at those coordinates. Random spot-level splits leak, since
 121 a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous
 122 blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

123 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
 124 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
 125 et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit
 126 neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods
 127 were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance
 128 read-out head — the most favorable choice available — and is marked *-style*, since these are not
 129 reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and
 130 evaluation code.

131 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,
 132 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because
 133 correlation and RMSE both reward regression toward a smooth mean.

134 **Two scopes, kept distinct.** Results are reported at two scales and every number is labeled with
 135 the one it belongs to. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and is the
 136 basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full
 137 budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher
 138 on the primary section, and a value from one is never quoted in support of a claim about the other.

139 4 Results

140 4.1 Masked spatial reconstruction across 22 sections

141 We evaluate on 22 sections spanning four technologies (336 runs), with the section as the unit of
 142 analysis: every model sees identical held-out blocks, so a paired test removes the between-section
 143 variance that otherwise dominates. NRDO attains Pearson r 0.183 against 0.170 for the closest
 144 baseline (STAGATE-style).

145 All 7 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix ??), and the *smallest* effect
 146 size across the family is $d_z = 0.80$, which is large by conventional standards; against every com-
 147 parator NRDO wins on at least 17/22 of the sections both were run on. The advantage is therefore
 148 consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span
 149 the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the
 150 variation between them is anatomical.

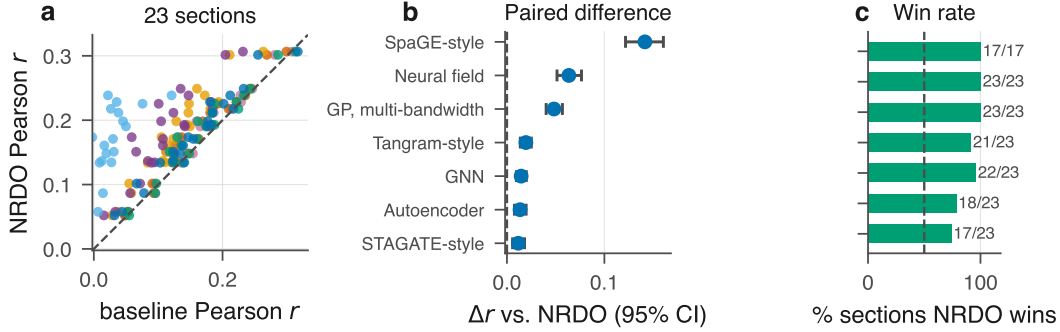


Figure 2: **Benchmark across 22 tissue sections.** (a) NRDO against each baseline, one point per section. (b) Mean paired difference with a percentile bootstrap 95% CI. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix ??.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.169 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to 0.002 Pearson r , against 0.014 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms (Section 5).

Spatial autocorrelation. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.335 against 0.347), and it has by some margin the largest error in Moran’s I (0.848 against 0.612 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

5 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (−0.023), and removing the reaction gives 0.241 ± 0.003 (−0.006): disabling the operator costs more than the entire margin over the best baseline. The control is exactly parameter-matched — the operator holds 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused — so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the control separates the two explanations. The full sweep, including the biological regularizers and the PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in Appendix ??.

6 Limitations

Identifiability. A single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather

than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport; Section 4 shows they are moreover close to their initialization, which is why we make no claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

Over-smoothing. The predicted fields are smoother than the measurements, and on Moran’s I error NRDO trails the baselines it leads on the other metrics. This survives training to convergence and is only partly correctable by a spectral-matching term (Section ??), so it bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so.

One baseline remains unresolved. The benchmark’s statistical resolution is set by independent specimens, not sections: twelve of 22 sections are serial sections of one brain. An earlier version of this work reported five specimens, at which no family-wise significant result was reachable at any effect size. We therefore added tissues rather than sections, reaching 10, and the separation now survives Holm correction against 3 baselines including GNN (GCN) ($p = 0.020$, 9/10 specimens). It does not against STAGATE-style (7/10, $p = 0.43$), whose attention-weighted neighbourhood aggregation is the closest existing analogue to what the operator does. We take that as the honest boundary of the claim: the continuous formulation is distinguishable from most of this literature and not from all of it. One section (visium_human_heart) is excluded from every specimen-level analysis because its held-out set falls below 200 locations, at which Pearson r cannot be estimated to the precision the comparison requires.

Transfer. Under the held-out-block protocol the fitted operator does not transfer to a new tissue at all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline transfers better. It is a within-section model, and we no longer claim otherwise.

Scope. The *-style* baselines implement each method’s architectural core with a common read-out head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget and subsampled sections so that 336 runs were feasible on CPU, and scope is limited to single 2-D sections.

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A Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 1 (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every sufficiently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (4)$$

where $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ , and $R_{\Delta t}$ does not depend on the observed data.

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. $\square$

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table ?? (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

B Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section ??.

Theorem 2 (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (5)$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

317 **Corollary 3** (What a stationarity fit can determine). *Under the hypotheses of Theorem 2, the steady-*
 318 *state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about*
 319 *f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the*
 320 *observed manifold—is determined by the inductive bias of the architecture and the regularizers, not*
 321 *by the data.*

322 Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section ?? dis-
 323 places $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states
 324 off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is
 325 therefore consistent with the theory rather than merely an empirical disappointment: a single sta-
 326 tionary snapshot does not contain the information such a counterfactual would require. Breaking the
 327 degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal
 328 sampling, or interventional data.